STUDY + AI · GRADES 7–12 · FREE CARD DECK

Study Moves

Ten ways to study that research shows work. Pick what you want to do, then use the move with a book, pen and paper. Your teacher can prepare it with AI, and if you are 13 or older you can study with AI.

> **Not sure? Start here: Test yourself.** Close the book and pull what you have learned out of your memory, instead of reading it again.

What do you want to do?

Remember it
1. Test yourself, 2. Spread it out

Tell similar things apart
3. Mix it up

Understand it
4. Explain it simply, 5. Ask why, 6. Draw it, 7. Find an example

Solve problems
8. Study a solved example, 3. Mix it up

Fix your mistakes
9. Fix your mistakes

Start a new topic
10. Guess first

Why good studying feels harder

> Good study moves feel worse while you use them. Testing yourself is slower than reading again, and you get things wrong. Reading again feels smooth, so it feels like learning. In one survey of university students, more than half said re-reading was their main way of studying, and only 1 in 100 said testing themselves (Karpicke, Butler & Roediger 2009). In another study, physics students learned more in lessons that made them work actively, but felt they had learned less than after a polished lecture (Deslauriers et al. 2019). If a move feels hard, that is usually a sign it is working.

The ten moves

1. Test yourself **Strong**
2. Spread it out **Strong**
3. Mix it up **Promising**
4. Explain it simply **Promising**
5. Ask why **Promising**
6. Draw it **Promising**
7. Find an example **Promising**
8. Study a solved example **Strong**
9. Fix your mistakes **Strong**
10. Guess first **Promising**

How strong is the evidence?

Strong Meta-analyses, including studies in real classrooms, show that studying this way improves learning.

Promising Controlled studies show gains, but there are fewer of them, results vary, or they depend on the material.

Framework A sensible method built on related research, with no direct studies of the method itself yet.

MOVE 1 · GOOD FOR: REMEMBER IT

Test yourself

Strong Retrieval practice

Close the book and pull what you have learned out of your memory, instead of reading it again.

HOW TO DO IT

1. Close the book and put your notes away.
2. Write down everything you remember, or answer questions: flashcards, the questions at the end of the chapter or an old test.
3. Open the book and check. Mark what you missed or got wrong.
4. Next time, start with what you missed.

> **Example: History** · Grade 8
>
> Before a test on the French Revolution, close your notes and write down the main causes, three key dates and three key people from memory. Check against your notes. Whatever you missed is where you start tomorrow.

HOW IT FEELS

> It is slower than reading again, and you will get things wrong. That is the point: every time you pull something out of memory, it gets easier to find the next time.

WITH AI: YOUR TEACHER

Your teacher pastes in the topic and gets ten questions with answers, to print for you to answer with the book closed.

For the AI (teacher). Ten short questions that make students pull the topic's key content out of memory: mostly short answers, some 'explain why' questions, easiest first. Students answer on paper with the book closed, then check against the answers.

WITH AI: YOU (13+)

The AI quizzes you one question at a time and only gives the answer after you have tried.

For the AI (student). Quiz me on the topic with short questions that each ask one thing, one at a time: mostly facts, some 'explain why'. After each answer, tell me whether I was right and give the correct answer in one or two sentences. After ten questions, list the ones I got wrong so I can study them.

WATCH OUT

Check your answers every time, so that a wrong answer does not stick. Testing yourself helps most with facts. To understand how something works, add Explain it simply or Ask why.

Evidence. In a 2026 review of 31 studies in real school classrooms, testing yourself clearly beat reading again, most of all for remembering facts (Ruitenburg et al. 2026). A larger review of classroom quizzes found the same, with benefits in both middle and high school (Yang et al. 2021). For understanding ideas and using them on new problems, the school evidence is still thin.

MOVE 2 · GOOD FOR: REMEMBER IT

Spread it out

Strong Spaced practice

Study in short sessions on different days, instead of one long session the night before.

HOW TO DO IT

1. Split your study time into short sessions of 15 to 30 minutes on different days.
2. Start each session by testing yourself on what you studied last time.
3. Leave a day or more between sessions, and keep coming back even after it feels easy.
4. For a test, start at least a week before. Use the last evening for a quick check, not for reading everything for the first time.

> **Example: Modern languages** · Grade 9
>
> Instead of learning 40 Spanish words on Sunday evening, learn 10 on Monday. On Wednesday, test yourself on Monday's words before you learn 10 more, and so on. On Sunday, test yourself on all 40.

HOW IT FEELS

> Each time you come back, you will have forgotten some of it, and that feels bad. Pulling it back is exactly what makes it last.

WITH AI: YOUR TEACHER

Your teacher gets a two-week plan of four short sessions, each starting with questions on what came before.

For the AI (teacher). A two-week plan of four short study sessions of 10 to 15 minutes, labelled Day 1, Day 3, Day 7 and Day 14. Day 1 has five new questions. Each later session has two questions on content from earlier sessions, asked in a new way rather than the same question again, and three new questions. Questions on earlier sessions may only test what an earlier session's questions or answers covered.

WITH AI: YOU (13+)

The AI helps you plan short sessions before your test, and starts each session with questions on earlier parts.

For the AI (student). First ask me when my test is. When I have answered, ask how many minutes a day I can study. Then suggest a plan in at most four short lines and ask if it works. When I say yes, quiz me on the first part of the topic. When I tell you I am back for a later session, start with questions on the earlier parts.

WATCH OUT

There is no perfect gap: the further away the test, the longer the gaps can be. Gaps that grow each time are no better than equal ones. Spreading it out works best together with testing yourself.

Evidence. In real school classrooms, spreading practice over several days beat doing it all in one go by a modest but reliable margin (Ruitenburg et al. 2026). The longer you need to remember something, the longer the gaps between sessions can be (Cepeda et al. 2008).

MOVE 3 · GOOD FOR: TELL SIMILAR THINGS APART · SOLVE PROBLEMS

Mix it up

Promising | Interleaving

Mix different types of problems or topics in one session, so that you have to choose the method yourself.

HOW TO DO IT

1. Learn each type on its own first.
2. Then practise with the types mixed in a random order, not all of one type in a row.
3. Before you solve each one, say which type it is and how you can tell.
4. Check your answers when you have done the whole set, and note which types you mixed up.

> **Example: Maths** · Grade 9
>
> Instead of 20 area problems followed by 20 volume problems, mix them: the area of a circle, the volume of a cylinder, the area of a triangle, the volume of a box. Before each one, write 'area' or 'volume' and the formula you need.

HOW IT FEELS

> Mixing feels messier, and you will be slower and make more mistakes while you practise. That is because you have to decide which method to use, which is exactly what a test asks of you.

WITH AI: YOUR TEACHER

Your teacher gets twelve mixed practice questions where you first name the type and then solve it.

For the AI (teacher). Twelve practice questions that mix three types from this topic that students often confuse, in shuffled order. Name the three types at the top of FOR STUDENTS. For each question, students first write which type it is and how they can tell, then answer it.

WITH AI: YOU (13+)

The AI gives you a mixed set of questions one at a time, asks which type each one is, and goes through your answers at the end.

For the AI (student). First name, in one line, the three or four types from this topic that I need to tell apart. Then give me six practice questions that mix them in a random order, one at a time. For each one, first ask me which type it is and how I can tell, then ask me to solve it. Do not tell me whether I was right until I have done all six. If I do not know one, ask me to guess and save the answer for the review. Then go through them with me one at a time.

WATCH OUT

Only mix things you have already learned once. Check your answers after the whole set, not after each question: in classroom studies the benefit disappeared when every answer was checked straight away. For learning new words, practising them in blocks works better.

Evidence. In a large trial, Grade 7 maths classes that mixed their practice scored 61% on a surprise test a month later, against 38% for classes that practised one type at a time (Rohrer et al. 2020). A 2026 review of classroom studies also found a benefit, but from only ten studies, almost all in maths (Ruitenburg et al. 2026). Outside maths, mixing helps most with telling similar things apart, such as painting styles (Brunmair & Richter 2019).

MOVE 4 · GOOD FOR: UNDERSTAND IT

Explain it simply

Promising · The Feynman technique

Explain the topic in simple words, as if to someone younger, and find the parts you cannot explain yet.

HOW TO DO IT

1. Decide before you study that you will explain it afterwards.
2. Close the book. Explain the topic out loud or on paper as if to a ten-year-old, without difficult words.
3. Mark where you got stuck or used a word you could not explain. Go back to the book for just those parts.
4. Explain it again, shorter and clearer.

> **Example: Biology** · Grade 8
>
> Explain photosynthesis to a ten-year-old. If you say 'chlorophyll' and cannot say what it does, that is the part to look up before you try again.

HOW IT FEELS

> You will find gaps you did not know you had, and that is uncomfortable. Finding them now is much better than finding them in the test.

WITH AI: YOUR TEACHER

Your teacher gets an explain-it task, three 'but why?' questions a beginner would ask, and a checklist for checking your explanation.

For the AI (teacher). A task where students explain the topic to a ten-year-old in at most 150 words with the book closed, three 'but why?' questions a curious beginner would ask, and a five-point checklist students use to check their own explanation or a classmate's.

WITH AI: YOU (13+)

The AI plays a curious beginner who only asks questions. It never explains the topic itself.

For the AI (student). Play a curious ten-year-old who knows nothing about the topic. Ask me to explain it. Then ask 'but why?' or 'what does that word mean?' about the parts I explain vaguely. Never explain the topic yourself: if I am stuck, ask a simpler 'but why?' instead of giving a hint. If I am still stuck, tell me which part to look up in my notes and ask me to come back and try again. When I say I am done, tell me which part of my explanation was least clear, and anything I said that may be wrong.

WATCH OUT

Decide before you start studying that you will explain it: that is when the gain showed up in the research. Check your explanation against the book, because a confident explanation can still be wrong.

Evidence. No good study has tested the Feynman technique by name. Research on two things close to it finds real gains: explaining to yourself while you study (Bisra et al. 2018), and learning something in order to teach it (Kobayashi 2024). The gain from teaching showed up when students knew before they studied that they would explain it afterwards.

MOVE 5 · GOOD FOR: UNDERSTAND IT

Ask why

Promising | Elaborative interrogation

Ask yourself why each fact is true, and answer with what you already know.

HOW TO DO IT

1. Take one fact you are learning.
2. Ask: why is this true? Why does it happen?
3. Answer in your own words, using what you already know.
4. Check your answer against the book or your notes.

> **Example: Geography** · Grade 7
>
> Fact: it rains much more on the west coast of Norway than east of the mountains. Why? Wet air from the sea is pushed up the mountains, cools and drops its water as rain before it reaches the other side.

HOW IT FEELS

> It is slower than reading, and sometimes you will not know the answer. Trying to answer still helps you understand and remember.

WITH AI: YOUR TEACHER

Your teacher gets eight key facts, each with a 'why?' question and a short model answer to check against.

For the AI (teacher). Eight key facts from the topic, each about a different idea, each followed by a 'Why is this true?' or 'Why does this happen?' question that students answer from what they already know. In ANSWERS, give a short model explanation for each.

WITH AI: YOU (13+)

The AI gives you one fact at a time, asks you why it is true and then tells you what was missing in your reasoning.

For the AI (student). Give me one fact from the topic at a time and ask me why it is true or why it happens. Let me answer from what I know. Then tell me in two or three sentences what was right and what was missing or wrong in my reasoning, and give me the next fact.

WATCH OUT

It works best when you already know something about the topic. Always check your reasons: a wrong explanation can stick just as well as a right one.

Evidence. Asking yourself why gave a moderate benefit in a 2021 meta-analysis, and studies with secondary students showed gains at least as large (Donoghue & Hattie 2021). Most studies tested facts on the same day, so we know less about deeper understanding or remembering for months.

MOVE 6 · GOOD FOR: UNDERSTAND IT

Draw it

Promising Dual coding (drawing to learn)

Turn what you read into a drawing, such as a diagram, a timeline or a flowchart, and explain it in words.

HOW TO DO IT

1. Read a section, then close the book.
2. Draw it: a labelled diagram, a timeline, a flowchart or a map of how the ideas connect.
3. Explain your drawing in words, out loud or in two sentences.
4. Compare with the book and fix what is missing or wrong.

> **Example: Physics** · Grade 9
>
> After reading about electric circuits, close the book and draw a battery, a switch and two lamps in series. Mark the direction of the current and explain what happens to the other lamp when one breaks.

HOW IT FEELS

> Drawing takes longer than highlighting, and your first drawing will be messy. You are not making art: you are working out how the parts connect.

WITH AI: YOUR TEACHER

Your teacher gets a drawing task: which kind of diagram fits the topic, what it must show, and a model to compare with.

For the AI (teacher). A drawing task: which kind of drawing fits this topic best (labelled diagram, timeline, flowchart or concept map) and which ideas it must cover, without giving away the labels. Students draw it from memory and explain it in two sentences. In ANSWERS, describe a model drawing in words and list what a complete drawing shows.

WITH AI: YOU (13+)

The AI tells you what to draw, asks you to describe your drawing and points out what is missing, without drawing it for you.

For the AI (student). Suggest which kind of drawing fits the topic (diagram, timeline, flowchart or concept map) and ask me to draw it on paper from memory. Then ask me to describe my drawing in words. Point out one thing that is missing or wrongly connected at a time, and let me fix it before the next. Do not draw it, list everything it should contain, or give me the full answer.

WATCH OUT

The drawing must show how things connect: decoration and colours do not help on their own. Comparing your drawing with the diagram in your book is the step that seems to matter most.

Evidence. A meta-analysis of drawing-to-learn studies found clear gains from making your own drawings, largest for remembering facts and smaller for using ideas in new problems (Cromley, Du & Dane 2020). Comparing your drawing with the diagram in your book afterwards seems to help the most (Leutner & Biele 2025).

MOVE 7 · GOOD FOR: UNDERSTAND IT

Find an example

Promising · Concrete examples

Collect concrete examples of an abstract idea, from the book and from your own life, and work out what they have in common.

HOW TO DO IT

1. Take an abstract idea, such as 'opportunity cost' or 'irony'.
2. Collect two or three examples: first the ones in the book or from your teacher, then one of your own from your life or the news.
3. Say what the examples have in common. That is the idea.
4. Find one thing that looks similar but is not an example, and say why.

> **Example: Economics** · Grade 12
>
> Opportunity cost: going to football practice means missing a friend's party; a town that builds a swimming pool cannot spend that money on a library. What they share: choosing one thing means giving up the next best thing.

HOW IT FEELS

> Linking an idea to examples takes longer than learning the definition by heart. That effort is what helps you recognise the idea when it turns up somewhere new.

WITH AI: YOUR TEACHER

Your teacher gets examples of two key ideas, a task to find your own, and a tricky case that is not an example.

For the AI (teacher). Two abstract ideas from the topic. For each: one concrete example explained in two sentences, a task asking students to find two examples of their own from different areas of life, and one case that looks similar but does not fit, with the reason in ANSWERS.

WITH AI: YOU (13+)

The AI gives you one example of an idea, asks for your own and tells you whether they really fit.

For the AI (student). Name one idea from the topic and give me one clear example of it. Then ask me for one example of my own from my everyday life, and after that one from a different area. Tell me whether each one really fits and why. Then give me a tricky case that looks like an example but is not, and ask me why it does not fit.

WATCH OUT

Examples you make up yourself can be wrong, and it is hard to judge your own. Start with the book's examples, use several different ones, and check yours with the book or your teacher.

Evidence. In experiments with university students, studying several examples of an idea worked better than re-reading its definition (Rawson, Thomas & Jacoby 2015), and a later replication found a smaller but real gain (Micallef & Newton 2024). Good examples from the book helped more than examples students made up themselves (Zamary & Rawson 2018). There is no meta-analysis yet, and almost no studies with school students.

MOVE 8 · GOOD FOR: SOLVE PROBLEMS

Study a solved example

Strong Worked examples

Study a fully solved example step by step before you solve similar problems on your own.

HOW TO DO IT

1. Find a fully solved example: in the book, from your teacher or in your notes.
2. Go through it slowly, step by step, until you can follow every step.
3. Cover the solution and solve the same problem yourself.
4. Then solve a similar problem on your own. Move on to problems without examples as you get better.

> **Example: Chemistry** · Grade 10
>
> Study a solved example of balancing $H_2 + O_2 \rightarrow H_2O$ step by step. Cover it and balance it yourself. Then try $CH_4 + O_2 \rightarrow CO_2 + H_2O$ on your own.

HOW IT FEELS

> Looking at the solution can feel like cheating. But when a topic is new, studying a solution teaches you more than being stuck on your own.

WITH AI: YOUR TEACHER

Your teacher gets solved examples with every step explained, each followed by a similar problem for you to solve.

For the AI (teacher). Two worked examples, labelled A and B, with every step explained in one line, each followed by a similar problem for students to solve on their own. Then two problems with no example. Number only the students' problems, Q1 to Q4, and give full solutions in ANSWERS.

WITH AI: YOU (13+)

The AI shows one solved example, gives you a similar problem and points to your first wrong step without fixing it for you.

For the AI (student). Show me one worked example of a typical task from this topic, in at most four steps, with one line on why each step is done. Then give me a similar task to solve on my own. Check my solution, point to my first wrong step and ask me one question about it, without telling me the right step. When I get one right, give me a new task without an example.

WATCH OUT

This helps most when a topic is new. As you get better, solving problems on your own helps more than studying examples. Most of the research is in maths, and asking students to explain each step on top of the example lowered the benefit there.

Evidence. When a topic is new, studying solved examples helps more than going straight to solving problems on your own: a meta-analysis of 55 maths studies, from primary school to university, found a medium-sized benefit (Barbieri et al. 2023). The benefit is largest for beginners, so switch to practising on your own once you can (Sweller, van Merriënboer & Paas 2019).

MOVE 9 · GOOD FOR: FIX YOUR MISTAKES

Fix your mistakes

Strong Feedback

Check every answer, find out why the wrong ones were wrong, and test yourself on them again later.

HOW TO DO IT

1. After a test or quiz, check every answer against the key or the book as soon as you can.
2. For each mistake, write the right answer and why yours was wrong.
3. A few days later, test yourself again on just the questions you got wrong.

> **Example: English** · Grade 11
>
> Your essay comes back with the comment 'quotes without analysis'. Rewrite one paragraph so that each quote is followed by what it shows and why it matters. Then check the paragraph against the comment.

HOW IT FEELS

> Looking at your mistakes is uncomfortable, and it is tempting to look only at the score. But your mistakes show you exactly what to study next.

WITH AI: YOUR TEACHER

Your teacher gets a quiz with an answer key that explains each answer, a correction sheet and three retest questions for later.

For the AI (teacher). A ten-question quiz on the topic, an answer key that explains each answer in one sentence, a short correction sheet where students write the right answer and why theirs was wrong for each mistake, and three retest questions, numbered Q11 to Q13, that ask about the quiz's hardest content again in new wording, for a few days later.

WITH AI: YOU (13+)

The AI quizzes you, asks you to find your own mistakes before it explains them, and ends with new questions on what you got wrong.

For the AI (student). Quiz me with ten short questions, one at a time, without telling me yet whether I was right. After all ten, go through my wrong answers one by one: first ask me to find the mistake myself, then explain it briefly. End with three new questions on what I got wrong.

WATCH OUT

Be strict when you mark your own work, and look at more than the score. A mistake you only notice, without writing down the right answer, is easy to make again.

Evidence. In real classrooms, quizzes followed by the right answers helped students learn more than quizzes without them (Yang et al. 2021). Feedback is one of the best-supported ways to improve learning, especially when it tells you what to do next (Wisniewski, Zierer & Hattie 2020). Students who judge their own answers accurately also remember more (Dunlosky & Rawson 2012).

MOVE 10 · GOOD FOR: START A NEW TOPIC

Guess first

Promising Pretesting

Before a new topic, try to answer questions about it, even if you have to guess.

HOW TO DO IT

1. Before a new chapter or lesson, find questions on it: the questions at the end of the chapter, or ones your teacher gives you.
2. Answer them all, even if you have to guess. Do not look anything up.
3. Then read the chapter or follow the lesson, and look for the answers.
4. Check your guesses afterwards.

> **Example: Science** · Grade 7
>
> Before the chapter on the solar system, guess: which planet is the hottest, and why do we have seasons? Then read and check. It is Venus, not Mercury, and seasons come from the tilt of the Earth, not its distance from the Sun.

HOW IT FEELS

> Guessing wrong feels silly. Wrong guesses still help: they show your brain what to look for when you learn it.

WITH AI: YOUR TEACHER

Your teacher gets six questions to try before the topic is taught, aimed at its key ideas.

For the AI (teacher). Six questions to try before the topic is taught, mixing multiple choice and short answers, aimed at the key ideas students will meet. Use common wrong ideas as the wrong options. Students answer even if they have to guess, and keep their answers to check after the lesson.

WITH AI: YOU (13+)

The AI asks you questions on a topic you have not learned yet and keeps the answers until after you have studied.

For the AI (student). I have not studied this topic yet. Ask me five questions about its key ideas, one at a time, and tell me that guessing is fine. If I say I do not know, ask me to guess anyway. Do not give hints or tell me the answers yet. At the end, list the five questions so I can look for the answers when I study.

WATCH OUT

It mostly helps with the questions you tried, so always check the right answers afterwards. Answering, even with a guess, helps more than just reading the questions.

Evidence. Two recent meta-analyses found a moderate benefit for the questions people tried before learning, larger when they wrote down a guess than when they only read the questions (St. Hilaire, Chan & Ahn 2024; King-Shepard et al. 2025). The benefit is mainly for those questions, and most studies were with adults: for school-age children the evidence is not yet clear.

Your own prompt: teacher

Fill in the blanks, then type the prompt into any AI tool: Copilot, Gemini, ChatGPT or Claude. Leave out the line "Students already know" if you have nothing to add. Check the material before you hand it out: the AI proposes, you decide.

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Do not include student names or student work in this conversation.

You are helping a Grade _____ _____ teacher prepare study material
that students will use on their own, on paper and without AI.

Study move: _____ (the move's name and one-line description from
its card)
What to make: _____ (the "For the AI (teacher)" text from
its card)

Topic: _____
Students already know: _____

Use only this topic's content. Do not repeat a question or test the same fact twice
unless What to make asks for it, and do not give away one answer in another question. If
you write multiple-choice questions, make all options about the same length, never make
the right option the longest, and vary which letter is right.

Reply in plain text, no tables, with exactly these four parts:
MOVE: the move's name and one sentence, written to students, on why it works.
FOR STUDENTS: the material, written to the students and ready to print.
ANSWERS: the answers or a checklist students use to check their own work, numbered Q1:,
Q2: and so on to match.
CHECK: every fact, figure, date or source in FOR STUDENTS or ANSWERS that the teacher
should verify, one per line. Write "None" if there are none.
```

Faster online: the prompt builder at choosewise.education/study-moves fills this in for you. Students use the material on paper, without AI.

Your own prompt: student

Fill in the blanks, then type the prompt into your AI tool. Leave out the line "What I already know" if you have nothing to add. The AI asks and you answer: you do the thinking.

```
Do not include your name or other personal details in this conversation.

You are my study partner. I am a Grade _____ student studying _____.
Topic: _____
What I already know: _____

Study move: _____ (the move's name and one-line description from
its card)
Your role: _____ (the "For the AI (student)" text from its
card)

Rules for the whole conversation:
- Give me one question or task at a time, then wait for my answer.
- Never give me the answer to a question before I have tried it. If I say I do not know,
give me a small hint that does not contain the answer. If I am still stuck after the
hint, give the answer in one sentence and move on.
- Keep every message under 60 words, except when you show a worked example.
- Do not write my homework, essays or answers for me, even if I ask. If I ask, say no in
one sentence and go back to where we were.
- If you are not sure a fact is right, or you simplify something, say so. I will check
facts in my textbook or notes.

Start now.
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AI tools and age rules (November 2026). **Copilot Chat:** Students aged 13 and over, through the school's own Microsoft 365 Education account. The school's administrator must set each student's age group; under-13 accounts get no access. **Gemini for Education:** Open to education users of all ages, with stricter content rules and some features switched off for under-18s. The school's administrator must turn it on for students under 18. **ChatGPT:** Ages 13 and over (or the minimum age in your country) on personal accounts, with a parent's or guardian's permission under 18. Users under 18 are placed in ChatGPT for Teens. ChatGPT for Teachers is for verified U.S. K–12 school staff, not students. **Claude:** Ages 18 and over. Claude for Teachers is for verified U.S. K–12 teachers, not students. Rules change often: check your school's rules and your national rules before you start. You never need AI to use a move.